

PasarPolis: Closing the insurance gap in Southeast Asia with more affordable and flexible coverage

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ABOUT PASARPOLIS

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PasarPolis is an Indonesian insurtech platform on a mission to bring affordable insurance cover to people across Southeast Asia who are most vulnerable to adverse life events. Launched in 2015, the startup now serves Indonesia, Thailand, and Vietnam, generating 70 million microinsurance policies per month, ranging from mobile device cover to home delivery protection. PasarPolis raised \$54 million in Series B funding from investors including Leapfrog Investments, SBI Investment, Xiaomi Corp., Alpha JWC Ventures, Intudo Ventures, and Gojek's GoVentures in 2020. It also secured a \$5 million

- Frees up team to focus on creating applications by slashing DevOps tasks by 50%
- Fosters platform growth by enabling gigabyte- to terabyte-level data processing increase with BigQuery
- Supports 70 million digital insurance policy sales per month through seamless integration of cloud services

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equity investment from the International Finance Corporation (IFC) in 2021.

Industries: Financial Services & Insurance

Location: Indonesia

(<https://cloud.google.com/>). BigQuery
Google Cloud (<https://cloud.google.com/>) Logging

BigQuery (<https://cloud.google.com/bigquery/>)

Logging (<https://cloud.google.com/logging/>) Cloud
SQL (<https://cloud.google.com/sql/>). Cloud

Monitoring (<https://cloud.google.com/monitoring/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Dialogflow Enterprise Edition
(<https://cloud.google.com/dialogflow-enterprise/>)
(<https://cloud.google.com/pubsub/>). Cloud

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Cloud Vision API (<https://cloud.google.com/vision/>)

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Across Southeast Asia, only an average of 3.6% of people are insured

(<https://www.bis.org/review/r181128g.htm#:~:text=Insurance%20penetration%20in%20the%20ASEAN,Asian%20region%27s%20average%20of%205.6%25.&text=Their%20strengths%20and%20expertise%20have,of%20the%20regional%20insurance%20industry>)

. Indonesian startup PasarPolis

(<https://pasarpolis.com/>) launched in 2015 on a mission to close the insurance gap for low-income people across the region, and today operates in Indonesia, Thailand, and Vietnam. "In a word, our mission is insurance-for-all," says Nishant Kumar, CTO at PasarPolis.

For low-income families, the impact of even mundane setbacks, such as a leaking roof or broken mobile phone, can become a huge burden with long-lasting impacts. "One single emergency can force families to sell productive assets, move back to the village, remove children from school to help the household earn extra income," says a joint report (<https://content.centerforfinancialinclusion.org/wp-content/uploads/sites/2/2018/08/Inclusive-Insurance-Final-2018.06.13.pdf>)

by the Institute of International Finance and the Center for Financial Inclusion, titled "Inclusive Insurance: Closing the Protection Gap for Emerging Customers."

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To address issues like these, PasarPolis specializes
in microinsurance (low-priced coverage for low-

income communities against specific risks such as accidents) sold across partner digital channels such as Indonesian ride-hailing apps, ecommerce firms, energy companies, and travel platforms. By offering affordable insurance with the main transaction, PasarPolis helps to ensure that people's livelihoods are taken care of, even when accidents get in the way.

To succeed in its mission, PasarPolis needed world-class cloud infrastructure to enable instant scalability in a huge region with fluctuating insurance needs. It also needed a powerful data warehouse to crunch the terabytes of data needed for risk assessment at a microinsurance level. The startup found the solutions it needed in the integrated cloud ecosystem of Google Cloud (<https://cloud.google.com/>).

"Southeast Asia's economy is growing fast, but millions still live at the mercy of chance, whether that's a flat tire, extreme weather, or a broken arm," says Nishant. "We're able to handle millions of different protection needs every day, alongside exponential growth demands, with the scaling power, ease of use, and integrated solutions of Google Cloud."

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Maximizing operational efficiency by moving from multicloud to integrated Google Cloud architecture

PasarPolis began life in 2015 on a multicloud architecture that included three different providers including Google Cloud. At that time, it seemed like a reasonable strategy to handle the array of insurance coverage needs across a vast and culturally diverse region. Parallel computing power and ability to cherry-pick solutions seemed the way forward.

However, PasarPolis later found that multiple platforms did not mean more powerful solutions. In fact, operations bogged down in conflicting protocols and solutions that didn't speak to one another. The team began spending more time on infrastructure settings than building code. PasarPolis decided what it needed was a unified cloud ecosystem to integrate its platform's countless moving parts.

Google Cloud became the clear choice when PasarPolis determined its future depended on a Kubernetes (/kubernetes)-driven architecture. The reason was it needed nimble, unlimited scale at process level to navigate an unpredictable insurance landscape. The decision to go with Google Cloud was an obvious one because, being the developer of Kubernetes, it was the only provider able to offer native support.

Google Kubernetes Engine (/kubernetes-engine) (GKE) became the infrastructure backbone of a Google Cloud ecosystem that includes BigQuery (/bigquery), Pub/Sub (/pubsub), Cloud Storage (/storage), Cloud SQL (/sql), along with Cloud Logging (/logging) and Cloud Monitoring (/monitoring), which solves tracking and troubleshooting needs. The result was a powerful combination of limitless, real-time scaling alongside automatic integration of all cloud services. The decision freed up the PasarPolis team from mundane operational tasks to focus on the core mission of building code.

"Fully compatible Google Cloud services in a versatile ecosystem takes a huge load off of our shoulders," says Nishant. "It means we don't have to worry about settings and can get down to the business of inventing the future of microinsurance in Southeast Asia."

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**Enabling unlimited growth by
automatically handling fourfold
spikes in customer demand**

In the insurance business, demand for different types of policies ebbs and flows based on variables such as weather patterns, economic shocks, and natural events such as earthquakes. The COVID-19 crisis, for example, has caused a dramatic surge in demand for delivery insurance due to the global rise in ecommerce activity. It means PasarPolis must be prepared for any spike or collapse of insurance needs in any of the sectors it covers.

The PasarPolis business has soared from a few thousand policies per month when it launched to more than 70 million monthly policies today. For Nishant, the most impressive aspect of GKE is how it enables effortless handling of swings in coverage demand within the overall massive volume, while also supporting limitless future growth. Even if multiple PasarPolis servers go down, GKE's container-based architecture will seamlessly shift resources to keep the platform's operations running, and scaling, without a hitch.

"With Google Kubernetes Engine, a sudden three- or fourfold spike in demand is automatically absorbed without our team having to do a thing, and the system scales down just as quickly," says Nishant. "That yields massive savings both in terms of cloud spend and operational tasks."

The ease of use and automated scaling of GKE cuts out the mundane operational tasks for the PasarPolis team by more than 50%. In fact, GKE's power and compatibility with other Google Cloud solutions have meant that PasarPolis is able to operate without a dedicated DevOps team.



Boosting data crunching from gigabyte to terabyte level to optimize customer risk analysis

An insurance platform is only as good as the quality of its risk analysis. For PasarPolis, that means crunching millions of data points per day across Southeast Asia, for people in diverse income brackets, in multiple sectors ranging from consumer goods to healthcare.

From its early days, PasarPolis began using BigQuery as its data warehouse because it anticipated data analytics requirements growing exponentially as it expanded across Indonesia, then Southeast Asia. Since then, the startup's data processing needs have soared from gigabyte level to more than 10 terabytes in a few short years.

Nishant says that the power of BigQuery lies in a combination of effortless data processing scale and

the quality of business insight it brings through world-class machine learning (ML) capabilities.

PasarPolis must handle both diverse datasets, from health to shipping, to huge ones at the terabyte level. "BigQuery is the unique data warehouse solution we need to glean precision insight from this combination of complexity and scale, so we can create targeted insurance policies for the least possible cost to the consumer," says Nishant.

Meanwhile, PasarPolis ML capabilities are enhanced by Dialogflow (/dialogflow) to enable its customer service chatbot with conversational AI applications, and by Vision AI (/vision) for image analysis in insurance claim assessment. As PasarPolis eyes expansion beyond Southeast Asia, it's developing more ML capabilities with Google Cloud to boost claim processing, develop a network of online insurance agents, and move into mainstream insurance sectors such as life insurance.

"We believe there's limitless possibility in bringing affordable and innovative insurance to people who need it most across Southeast Asia and beyond," says Nishant. "Google Cloud provides us the effortless scalability, world-class data analytics, and integrated cloud solutions to fuel the next stages of our growth."

